Mar. 12

- Remarks on notation:
  - $\in$ = "is an element of"
  - e = for us, usually denotes the identity of a group
  - ε = a small positive number
  - E = capital e (sets are often denoted with capital letters
  - $\exists$ = "there exists"
  - $\exists!$ = "there exists a unique"

- A function $\varphi: \overset{(G,*)}{G} \to \overset{(H,\star)}{H}$ is a <u>homomorphism</u> $\varphi(a*b) = \varphi(a) \star \varphi(b)$ $\forall a,b \in G$.
  - If φ is also bijective, then we say φ is an <u>isomorphism</u> and we write $G \cong H$

- 3 main types of objects in math: $\overset{G,H}{\text{sets}}$, $\overset{\varphi}{\text{functions}}$, $\overset{a,b}{\text{elements}}$
  - Be careful to keep track of what kind of objects each thing is

  <u>ex</u> $a*b$ ✓ $G*H$ ✗ watch out for <u>type errors</u> like this

- <u>Prop</u> If $\varphi: G \to H$ is a homomorphism, then $|\varphi(a)|$ divides $|a|$ $\forall a \in G$

  If φ is an isomorphism, then $|\varphi(a)| = |a|$

  for now we only prove the case for finite order

  <u>Proof</u> Suppose φ is a homomorphism and $|a| = n$.

  We see $a^n = e_G$, so $\varphi(a^n) = \varphi(a_G) = e_H$. further, $\varphi(a^n) = \varphi(\underbrace{a*\ldots*a}_{n \text{ times}}) = \varphi(a) \star \underbrace{\varphi(a*\ldots*a)}_{n-1 \text{ times}}$

  $= \underbrace{\varphi(a) \star \ldots \star \varphi(a)}_{n \text{ times}}$. Thus $\varphi(a^n) = \varphi(a)^n$, so $\varphi(a)^n = e_H$.

  By result in Ch. 3, $|\varphi(a)| = n$.

  If, in addition, φ is bijective, then write $|\varphi(a)| = k$. We see $n = k \cdot m$ for some $m \in \mathbb{Z}^{>0}$. Now $\varphi(a)^k = e_H$, and $\varphi(a^k) = \varphi(a)^k$. Thus $\varphi(a^k) = e_H$. Since φ is bijective and since $\varphi(e_G) = e_H$, we see $a^k = e_G$. By result in Ch. 3, $|a| = n$ divides k. Since $k|n$ and $n|k$, we see $k = n$. $\square$

<u>Ex</u> $\varphi: \mathbb{Z}/8\mathbb{Z} \to (\underbrace{\{i,-i,1,-1\}}_{\langle i \rangle \text{ under mult.}}, \cdot)$ given by $\varphi(\bar{1}) = i^1$. More generally, $\varphi(\bar{a}) = i^a$

Check if well-defined

$\bar{a} = \bar{b} \Leftrightarrow a = b + 8k$, $\varphi(\bar{a}) = i^a$ & $\varphi(\bar{b}) = i^b$

$i^{b+8k} = i^b i^{8k} = i^b \cdot 1^k = i^b$ ✓

Check if homomorphism left as exercise

$|\bar{1}| = 8$, $|\varphi(\bar{1})| = |i| = 4$, $4|8$ ✓

Assoc. of composition

- Given three functions $f: A \to B$, $g: B \to C$, $h: C \to D$, we can write $h \circ (g \circ f) \overset{?}{=} (h \circ g) \circ f$

  Let $x \in A$ and write $y = f(x) \in B$ and $z = g(y) \in C$. We compute $(h \circ (g \circ f))(x) = h \circ (g(f(x))) = h(g(y)) = h(z)$.

  Also compute $((h \circ g) \circ f)(x) = h(g(y)) = h(z)$. Thus, we have associativity of function comp.

  $(h \circ g)(f(x)) = (h \circ g)(y)$

Cosets and Lagrange's Theorem

Abstract Algebra

Let G be a group and H a subgroup of G. For any $a \in G$ define the *left coset of H in G determined by a* to be

$$aH = \{ah : h \in H\}.$$

We say that a is a *coset representative* for aH. There are usually multiple possible coset representatives for a given coset. The set of *left cosets of H in G* is $\{aH : a \in G\}$. We can define right cosets analogously.

1) (Ch.7 Ex. 1) Let $H = \{0, \pm 3, \pm 6, \pm 9, \dots\}$. Find all the left cosets of H in $\mathbb{Z}$. Find all the *right* cosets of H in $\mathbb{Z}$. Do you think left and right cosets are going to be that different in Abelian groups?

$H = 3\mathbb{Z} < (\mathbb{Z},+)$, so $aH = a+H$. $1+H = 1+3\mathbb{Z} = \{\dots,-5,-2,1,4,\dots\} = \bar{1}_3$, $2+3\mathbb{Z} = \bar{2}_3$, $0+3\mathbb{Z} = \bar{0}_3$, $4+3\mathbb{Z} = \bar{4}_3 = \bar{1}_3$, $5+3\mathbb{Z} = \bar{5}_3 = \bar{2}_3$. The set of cosets is $\{\bar{0}_3, \bar{1}_3, \bar{2}_3\} = \mathbb{Z}/3\mathbb{Z}$

left right

Left and right cosets are probably the same in Abelian groups, since in such a group, $ah = ha \ \forall a,h \in H$

2) (Ch.7 Ex. 3) Decide whether the following cosets of $H = \{0, \pm 3, \pm 6, \pm 9, \dots\}$ in $\mathbb{Z}$ are the same:

$H = 3\mathbb{Z}$

a. $11 + H$ and $17 + H$. $11+3\mathbb{Z} = \bar{11}_3 = \bar{2}_3$, $17+3\mathbb{Z} = \bar{17}_3 = \bar{2}_3$ $\quad 11+H = 17+H$ ✓

b. $-1 + H$ and $5 + H$. $-1+3\mathbb{Z} = \bar{-1}_3 = \bar{2}_3$, $5+3\mathbb{Z} = \bar{5}_3 = \bar{2}_3$ $\quad -1+H = 5+H$ ✓

c. $7 + H$ and $23 + H$. $7+3\mathbb{Z} = \bar{7}_3 = \bar{1}_3$, $23+3\mathbb{Z} = \bar{23}_3 = \bar{2}_3$ $\quad 7+H \neq 23+H$ ✗

d. $H + 4$ and $-5 + H$. Since $(\mathbb{Z},+)$ is Abelian, $H+4 = 4+H = \bar{4}_3 = \bar{1}_3$
$-5+H = \bar{-5}_3 = \bar{1}_3$ $\quad H+4 = -5+H$ ✓

3) (Ch.7 Ex. 4) Let $n \in \mathbb{Z}^{>0}$ and $H = \langle n \rangle$. Find the left cosets of H in $\mathbb{Z}$. How many are there? Do you already have a name for them?

$H = \langle n \rangle = n\mathbb{Z} = \{0, \pm n, \pm 2n, \dots\}$

$0+H = \{0, 0, \dots\} = \{0\} = \bar{0}_n$

$1+H = \{1, 1+n, 1+(-n), \dots\} = \bar{1}_n$

The left cosets for n in $\mathbb{Z}$ are $\{\bar{0}_n, \bar{1}_n, \bar{2}_n, \dots, \overline{n-1}_n\} = \mathbb{Z}/n\mathbb{Z}$

There are n such cosets

4) Find all the left cosets of $\langle (12) \rangle$ in S_3. Write down some things you notice.

$\langle (12) \rangle = \{(12), (12)(12) = id\}$

$id\langle (12) \rangle = (12)\langle (12) \rangle = \langle (12) \rangle$

$id\langle (12) \rangle = \{(12), id\}$

$(13)\langle (12) \rangle = \{(13)(12) = (123), (13)\}$

$(12)\langle (12) \rangle = \{id, (12)\}$

$(123)\langle (12) \rangle = \{(123)(12) = (13), (123)\}$

$(23)\langle (12) \rangle = \{(23)(12) = (132), (23)\}$

$(132)\langle (12) \rangle = \{(132)(12) = (23), (132)\}$

equal, different coset representatives

$S_3 = \{id, (12), (13), (23), (123), (132)\}$

elements of a given coset are the possible representatives of the coset

Worksheet has a back side, didn't get to it in class, encouraged to work on it yourself over the weekend, preparing to prove Lagrange's Thm. next class